

Aggregate Functions in SQL

Q. What are Aggregate Functions in SQL? Explain the functions COUNT(), SUM(), AVG(), MAX(), and MIN() with syntax and appropriate examples.

 **Definition to Remember**

Aggregate Functions in SQL operate on a set of values (an entire column) and return a **single summarizing value**. They are used for data analysis and reporting, typically combined with the **GROUP BY** clause. With the exception of **COUNT(*)**, **all aggregate functions ignore NULL values**.

The Five Standard Aggregate Functions

Function	Returns	Works On
COUNT()	Number of rows	Any type
SUM()	Total sum	Numeric only
AVG()	Arithmetic mean	Numeric only
MAX()	Highest value	Numeric, Text, Date
MIN()	Lowest value	Numeric, Text, Date

1. COUNT()

Returns the total number of rows matching a criterion.

- * **COUNT(*)** → Counts **all rows** (including NULLs).
- * **COUNT(column)** → Counts rows where that column is **NOT NULL**.

```
SELECT COUNT(*) FROM Employee;           -- total rows
SELECT COUNT(Manager_ID) FROM Employee;  -- rows where Manager_ID is not null
SELECT COUNT(DISTINCT Department) FROM Employee; -- unique departments
```

2. SUM()

Calculates the total sum of a numeric column.

```
SELECT SUM(Salary) FROM Employee;  -- total monthly salary payout
```

3. AVG()

Calculates the arithmetic mean of a numeric column.

```
SELECT AVG(Salary) FROM Employee WHERE Department = 'IT';
```

4. MAX()

Returns the maximum (highest) value. Works on numbers, text (alphabetically), and dates.

```
SELECT MAX(Salary) FROM Employee; -- highest salary
SELECT MAX(Name) FROM Employee; -- last name alphabetically
```

5. MIN()

Returns the minimum (lowest) value.

```
SELECT MIN(Salary) FROM Employee; -- lowest salary
```

NULL Behavior in Aggregate Functions

- `SUM()`, `AVG()`, `MAX()`, `MIN()` — ignore NULL values.
- `COUNT(*)` — counts all rows including those with NULLs.
- **Example:** Salaries = [100, 200, NULL] → `AVG(Salary)` = (100+200)/2 = **150** (NULL ignored).

★ Must-Write Points (for 10 marks)

1. Aggregate functions compute a single value from a set of values (a column).
2. `COUNT(*)` counts all rows including NULLs; `COUNT(column)` excludes NULLs.
3. `SUM()` and `AVG()` work on **numeric** columns only.
4. `MAX()` and `MIN()` work on numeric, text (alphabetical), and date values.
5. All aggregate functions **ignore NULL values** — except `COUNT(*)`.
6. Use `DISTINCT` inside aggregate functions to calculate on unique values only.
7. Aggregate functions are most powerful when combined with `GROUP BY` and `HAVING`.

⚡ Quick Recall

Aggregate Functions → `COUNT(*)` / `COUNT(col)` → `SUM (numeric)` → `AVG (numeric)` → `MAX / MIN (any type)` →
Ignore NULLs (except `COUNT(*)`) → Used with `GROUP BY`